NOTE

However, in practice the arithmetic mean is more frequently used in calculating the mean deviation. If specifically stated to calculate mean deviation about median, median can be used.

Example 8.19

Calculate the Mean Deviation about mean and its coefficient of the income groups of five, given below.

Income Rs.	4000	4200	4400	4600	4800

Solution:

Calculation of Mean Deviation about Mean

$$\text{Mean} = \frac{\sum X}{n} = \frac{22000}{5} = 4400$$

Income (Rs) X	$\lvert D\rvert = (X - 4400)$
4000	400
4200	200
4400	0
4600	200
4800	400
$\sum X = 22000$	$\sum \lvert D\rvert = 1200$

Table : 8.13

$$\text{Median Deviation about Mean } MD = \frac{\Sigma |D|}{n}$$

$$\text{MD} = \frac{1200}{5} = 240$$

$$\text{Coefficient of } MD = \frac{240}{4400} = 0.055$$

Example 8.20

Calculate the mean deviation about median and its relative measure for seven numbers given below: 55, 45, 40, 20, 60, 80, and 30.

Solution:

Arrange the values in ascending order 20, 30, 40, 45, 55, 60, 80.

$$\text{Median} = \text{size of } \left(\frac{(n+1)}{2}\right)^{th} \text{ value when n is odd}$$

$$= \text{size of } \left(\frac{(7+1)}{2}\right)^{th} \text{ value}$$

$$= \text{size of } 4^{th} \text{ item} = 45$$

X	$\lvert X - \text{Median}\rvert = \lvert X - 45\rvert$
20	25
30	15
40	5
45	0
55	10
60	15
80	35
	$\Sigma \lvert X - \text{Median}\rvert = 105$

Table : 8.14

$$\text{MD about Median} = \frac{\sum |X - Median|}{n}$$

$$= \frac{105}{7} = 15.0$$

$$\text{Coefficient of MD about median} = \frac{15}{45} = 0.33$$

Example 8.21

Calculate the Mean deviation about mean for the following data.

Size	2	4	6	8	10	12	14	16
frequency	2	2	4	5	3	2	1	1

Solution:

Calculation of Mean Deviation about Mean

From the Table 8.15, we get

$$\overline{X} = \frac{\Sigma fX}{N} = \frac{160}{20} = 8$$

Mean Deviation about Mean

$$= \frac{\Sigma f |D|}{N} = \frac{56}{20} = 2.8$$

08_11th_BM-STAT_Ch-8-EM.indd 192 21-04-2020 12:26:32 PM